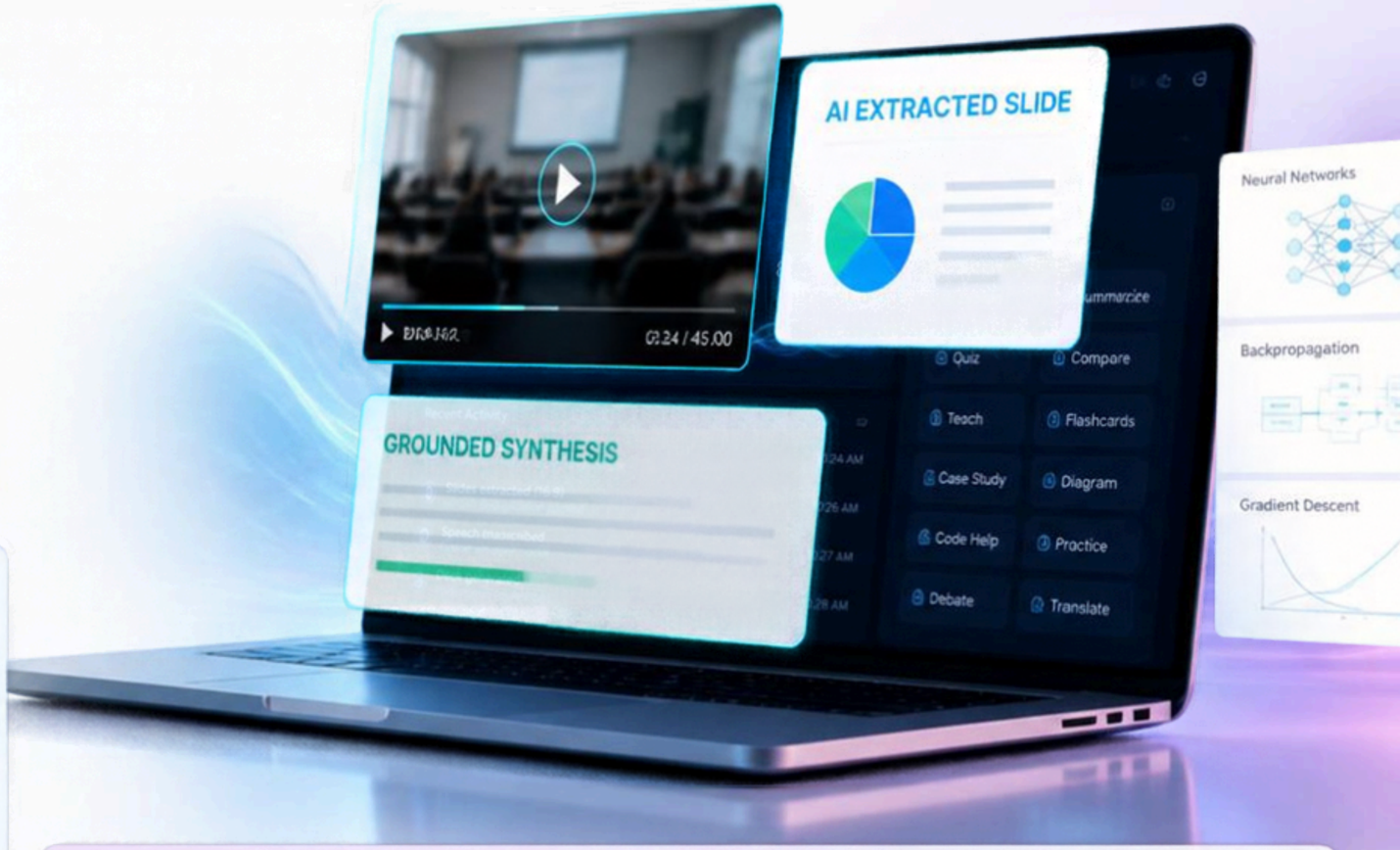
LearnLens AI

Autonomous Multimodal Knowledge Ingestion & Intelligent Learning Companion

Show your AI what you are learning.

LIVE IN PRODUCTION

Universal Web & Local-First AI



Autonomous multimodal AI agent

- Real-time Chrome tab observation and capture
- 16:9 slide extraction with dHash deduplication
- Speech transcription with faster-whisper
- Auto-generation of widescreen PowerPoint decks
- PDF guides and structured learning materials
- 12-mode interactive AI tutoring for any learner

Official Deployment & Repository Links

Live Web Application	transcript-agent-frontend.onrender.com
GitHub Repository	github.com/soham-092510/Transcript-Agent
FastAPI Swagger API	transcript-agent-backend.onrender.com/docs
System Status / Health	transcript-agent-backend.onrender.com/api/system/status

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The Enterprise Knowledge & Training Bottleneck

Why Digital Upskilling is Broken in Modern Engineering Organizations

100+

Hours of Video Fatigue

Engineers spend 40-100 hours watching lectures.
80% knowledge locked in videos, no search.

40%

Fragmented Manual Note-Taking

Pauses and note switching cause 40% drop in retention.
Diagrams lost or poorly saved.

Mid-Lifecycle

Knowledge Loss

Lack of unified verifiable materials tied to video timesstamps.

68% of learners cannot quickly find visual diagrams

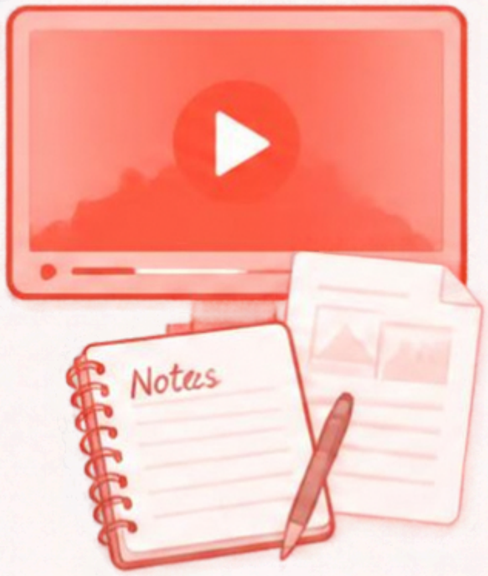
Zero searchability in video audio

LearnLens AI — The Autonomous Learning Companion

Passive Screen Observation Meets Active Multimodal Intelligence

Passive. Static. Manual.

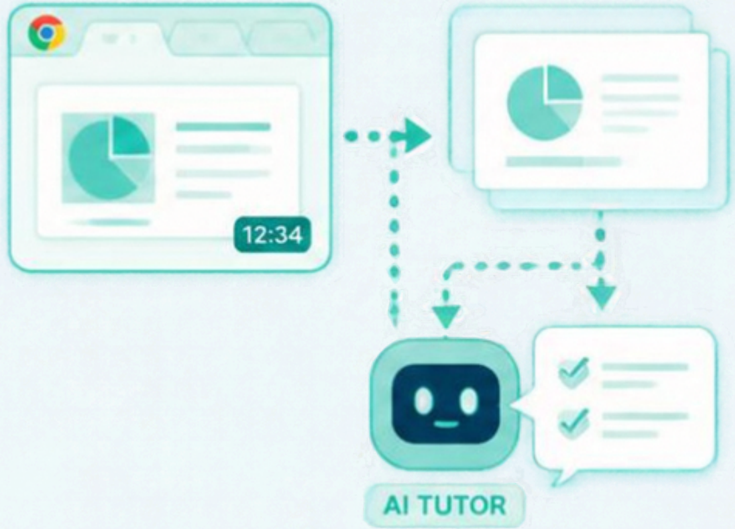
Legacy Video Learning



Manual notes and fragmented learning with no continuity

Observe. Extract. Teach.

LearnLens AI Agent



Autonomous capture, structured assets, and AI-powered learning

Zero-Lag

Chrome Tab Observation
<50ms startup

Intelligent

Keyframe Extraction
genuine slide transitions, no duplicates

Instant Study

Artifact Synthesis
16:9 PPTX & PDF decks

Grounded AI

Teacher & Quiz Engine
timestamped evidence & citations

System Architecture & Engineering Flow

Full-Duplex Multimodal Pipeline Orchestrated on Snapdragon NPU

TECH STACK

FRONTEND
React 19
TypeScript
Vite 8
Web Audio API
Web Speech API

BACKEND
FastAPI with async Websockets

VISION
Client-side dHash slide hashing + OpenCV filtering

AUDIO
Dual STT pipelines (client Web Speech + Faster-Whisper container)

OCR
Tesseract with asyncio concurrency

DOCUMENT SYNTHESIS
python-pptx, reportlab

IBM Bob 2.0 Orchestration Layer

Architecture planning • Automated endpoint generation • Testing & deployment specs

Live Interface & Observation Studio

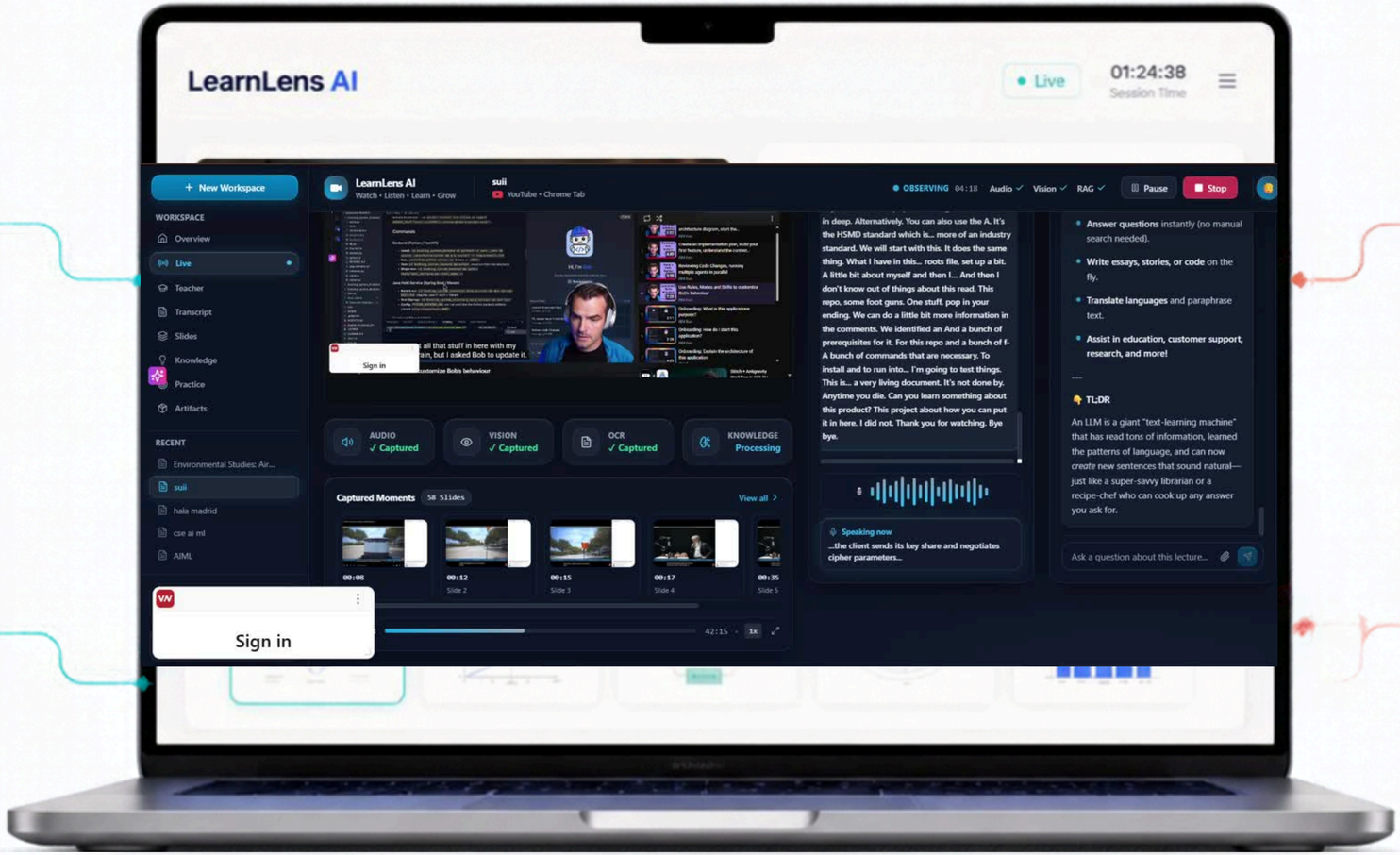
Production-Grade UI Engineered for Distraction-Free Learning

Sub-50ms Chrome Tab Observation

Zero-lag WebRTTC stream with PIP toggle

Autonomous Slide Change Carousel

Perceptual dHash extracts verified slides without duplicates



Real-Time Dual STT Feed

Sub-second speech recognition with live streaming ticker

1-Click 16:9 PPTX/PDF Export

Instant download of study decks and reports

HyperIngest Studio: 65x Accelerated Learning

Completing 11-Hour / 100-Video Master Courses Autonomously

Autonomous batch ingest of entire lecture directories / playlists

Non-linear frame scanning for slide extraction across 100+ videos

Chrome tab Auto-Pilot & Turbo Speed Script
2x-16x playback, auto Next Lecture on platforms

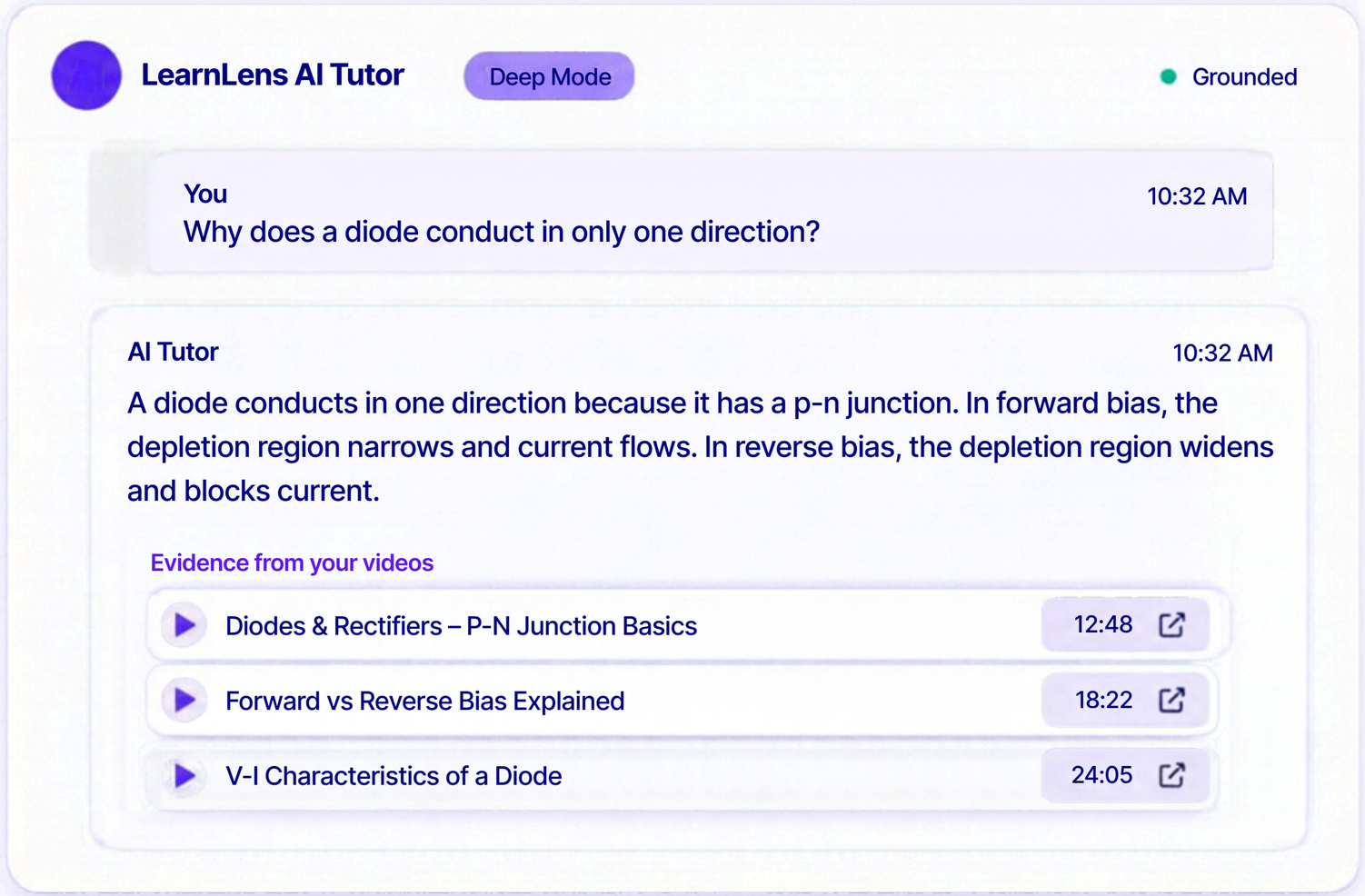
65x
Realtime
11-hour lecture ingested in under 12 minutes

85%
Network Savings
client-side change detection reduces video data sent

100%
Local Processing
no raw video uploaded, full privacy

Verifiable Learning & AI Doubt Solving

Eliminating Hallucinations Through Exact Slide & Timestamp Citations



Grounded & Accurate

Grounded Retrieval-Augumented Generation (RAG) ensures zero hallucination.

Three Teaching Modes

Simple mode for analogies
Deep mode for technical breakdowns
Exam Prep mode for high-yield summaries

Practice Exam Engine

Automated quiz generator with multiple-choice questions, instant grading, and timestamp evidence.

Production Engineering & Cloud Resilience

High-Throughput System Optimization for Scalable Multi-User Deployments

<0.2ms

Sub-millisecond SQLite queries

via WAL & B-tree indexing eliminating full scans

100%

Queue purging & CPU starvation safeguards

via thread-capped Whisper inference and chunk dropping

85%

Client-side deduplication & RMS silence gatling

cutting 85% redundant WebSocket data

100%

Cross-origin CDN resolution

dynamically routing frontend/backend asset requests

From Raw Concept to Deployed Production

Rapid 48-Hour Concept-to-Production Delivery Roadmap

01

Architecture & Prototyping

multi-tier specs
WebRTC capture
audio pipeline
DB schema

02

Vision & AI Synthesis Engine

perceptual dHash
Faster-Whisper workers
document generators

03

Hyperingest & Auto-Pilot Features

autonomous ingestion pipeline
auto-next automation
AI teacher

04

Cloud Deployment & Hardening

Render infrastructure as code
deployment
100% passing tests

Transforming Digital Education at Enterprise Scale

The Future of Multimodal Knowledge Ingestion is Here

IMPACT

Transforms

passive video into active, structured knowledge

60%+

saves study time with instant decks and quizzes

ROADMAP

- **Chrome Web Store** extension
- **LMS integration** (Canvas, Mood)
- **multi-lingual** translation

LIVE VERIFICATION LINKS

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GitHub Repository
github.com/soham-092510/Transcript-Agent

FastAPI Docs
transcript-agent-backend.onrender.com/docs

System Diagnostics
transcript-agent-backend.onrender.com/api/system/status

TEAM CONNECT

Let's build the future of learning together.

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